WavetableController Rev 4 — Final Red-Flag Report

Date	2026-10-13
Inputs	Rev-4 KiCad project (board with filled GND pour), BOM and Gerbers, sent by Trey 2026-10-13 (`rev4/`)
Baselines	Rev-3 review (`analysis/rev3/`, 65 entries) · 7 October "Rev 2 Design Review" rows · post-rev-3 chat decisions (LED values, /+9V_FLAG re-pointing, encoder RC, SD caps, inrush)
Scope	Passes 0–6 complete · 57 rev-4 ledger entries (`FINDINGS.md`) · every rev-3 finding re-scored · every Oct-7 row re-scored (`OCT7_REVIEW_STATUS.md`)
Truth	The rev-4 `.kicad_pcb` is copper truth; Trey's Gerbers agree with it on every number; nothing has been fabbed from rev 4
Verification	55 of 57 entries confirmed by script against the files; 1 plausible, waiting on Trey (pot taper); settled facts not re-opened

Decisions recorded after the passes ran (Kyle, 2026-10-13)

- The screen is **not mounted in this build** → Oct-7 row 5 is deferred.
- The LEDs are **hand-soldered Würth 150505M173300** (part already approved) → the missing BOM line is paperwork, not an assembly gap.
- The encoder filter caps **at the encoder end are accepted** (no room near the Seed; better than none).

Headline

Rev 4 is a targeted patch of rev 3, not a re-layout. Outside the changed parts the copper is identical to the millimetre: same 74.9 mm of 0.1 mm trace, same pour fragments, same SPI / SD / USB / MIDI / audio routes, same J8, same holes. Inside the patch Trey closed most of what was open after rev 3.

Oct-7 row	Rev 3	Rev 4
2c — U6 fed ahead of the 3.3 Ω filters	NOT DONE	**DONE** — LED/buck node ≈ 8.1 V, independent of the display
3c — LED resistors 510 Ω red / 390 Ω green-blue	NOT DONE	**DONE** — 11.6–12.1 mA, 57–69 mW in 0603
6b — L1 on the Murata DFE201610P pattern	NOT DONE	**DONE**
7 — Encoder 10 k pull-ups + 10 nF caps	NOT DONE	**DONE** — caps at the encoder end, accepted
9 — Delete D6; bridge to SOD-123FL	NOT DONE	**DONE** — 1N5817W, IFSM 25 A
SD-socket caps	NOT DONE	**DONE** — 10 µF at 5.2 mm, 100 nF at 6.9 mm from P1 VDD
5 — J8 position + four M2 holes	NOT DONE	**DEFERRED** — screen not mounted in this build
3d — LED part name / BOM line	DOCUMENTATION	**DOCUMENTATION** — hand-soldered Würth confirmed
Fab-margin rows (5) and tidy items	NOT DONE	**NOT DONE** — see Tiers 3 and 4

Nothing on the Oct-7 fix list blocks ordering rev 4. Of the 65 rev-3 findings: 4 fixed, 20 re-pass, 4 settled, 37 still open. None regressed; one got a pad worse (a ninth single-spoke GND thermal, on the new SD cap).

Tier 1 — Judgment items reviewed with Trey (nothing here is a fab rejection)

1. Grounding — withdrawn as an action after Trey's notes (2026-10-13)

(R4-P1-08, re-scoped twice and now closed; measurements on the rev-4 board)

- **Trey's first point, confirmed:** all parts are on the bottom; the bottom ground fill is one solid piece (12,789 mm², 77 % of the board); 73 % of signal routing is on top and returns through that plane. The top-layer fragments are in no return path, so stitching them is optional tidying, not a fix.
- **Trey's second point, confirmed:** U4 is a 0.4 mm-pitch DFN (pads 0.25 mm wide, 0.15 mm apart) and U6 a 0.5 mm-pitch WSON (0.25 mm pads, 0.25 mm apart). A 0.3–0.5 mm ground stub there would short to the neighbouring pin; 0.1 mm escapes are the only sane width between those pins, and a stub can only be pad-width (0.25 mm) where it leaves a pad along its own column. My suggested fix was wrong for these packages.
- **What the board actually does, measured:** every one of the three ICs has its **exposed pad on the plane** (U4.15, U6.9 with its two vias, U3.9), and the exposed pad is the real ground connection of each package. U4's ground pins 4 and 11 run straight into the exposed pad over 1.48 mm of 0.1 mm trace (≈ 7 mΩ); pins 7 and 9 reach it within 0.6–0.7 mm. U6's PGND/AGND/FB pins reach the pour over 1.05–1.63 mm; U3's two ground pins over 2.9 mm of 0.3 mm. These are the escapes the packages require, not a layout fault.
- **Potential consequence of leaving it:** none worth a change. The gain from re-routing U6's PGND inward to its exposed pad is a few milliohms and about a nanohenry.
- **Action:** none for this build. The nine single-spoke ground pads (C28, C9, C10, R18, U1.5, P2 A12/B1, TAC_SWITCH_1.3, U3.3) stay a Tier-4 hygiene note: set the zone's minimum spokes to 2 or shrink the thermal gap when convenient.

2. Inrush — no series resistor before C15; smaller than rev 3, but not limited

(R4-P2-05 · decision 5e not applied · first raised in the rev-3 review, not on the 7 Oct list)

- **The exact scenario:** the unit is off, so the 100 µF capacitor C15 holds no charge. The 9 V adapter is plugged in, or SW1 is flipped. For the first few microseconds an empty capacitor behaves like a short circuit, and the current that rushes in is limited only by what is in the path: the adapter's output, the barrel-jack contacts, the switch contacts, two diode drops and three ferrite beads at 0.05 Ω each. That is roughly a quarter of an ohm, so a peak of up to ≈ 30 A for ≈ 25 µs from a stiff adapter (an upper bound; a real adapter's output impedance lowers it).
- **What it does not hurt:** the diodes (the 1N5817W's 25 A rating is for an 8 ms pulse; this one is 300× shorter) and the adapter (it sags briefly).
- **Potential consequence:** the switch and jack contacts are the wear item. Each power-on puts a small spark across them as they close, and the slide switch does it several times per flip because of contact bounce. Over hundreds of cycles that pits the contacts; the symptom is a supply that drops out when the unit is bumped or the switch is wiggled. A slow failure, not a day-one one. Rev 1/2 never had this (10 µF at the input); rev 3 added 700 µF; rev 4 cut it to 310 µF, but the 100 µF directly on the unprotected path remains, and removing D6 and FB7 took 0.45 V and 50 mΩ out of the path.

- **Fix:** one part — a 1 Ω, 1 W resistor between FB5 and C15 (or an NTC). Peak drops to ≈ 8 A; cost 0.45 V and 0.2 W at full load; Daisy VIN still ≈ 7 V, inside 4–17 V. Can go in the next spin if not this one.

The only Tier-1 action left is the inrush resistor, and it can wait for the next spin.

Acknowledged / deferred by decision (Kyle, 2026-10-13)

Display mounting (Oct-7 row 5 · R4-P6-05)

- **Status:** "Acknowledged — will potentially deal with later." The screen is not mounted in this build.
- **What stands:** J8 is where it was; no M2 holes for the NHD module, no chassis holes for the board.
- **Potential consequence if the module is fitted later as-is:** it hangs on 20 solder joints behind the panel window until the holes exist; the enclosure window position stays open.

LED bookkeeping (R4-P4-01 · R4-P2-03 · R4-P6-04)

- **Status:** Trey hand-solders Würth 150505M173300; polarity and footprint are right for it. No assembly gap.
- **What stands:** schematic Value still "CLS6B-FKW" with the Cree pin labels; no hand-solder sheet.
- **Potential consequence:** the next person reading the files reaches the wrong part and the wrong polarity conclusion, as this review did in rev 3.
- **Fix (paperwork):** Value → 150505M173300, relabel the symbol pins, add a hand-solder sheet.

Encoder filter caps (Oct-7 row 7 · R4-P3-03 · R4-P5-04)

- **Status:** accepted as built — no room near the Seed.
- **What stands:** the four 10 nF caps sit at the encoders on the back, 6.8–10.2 mm from the encoder pins and 63–111 mm from the Daisy pins.
- **Potential consequence:** contact bounce is filtered at the source (good); the MCU input stays on an unterminated 63–111 mm line (noise pickup along the run). Keep the state-table quadrature decoder in firmware.

Tier 2 — Robustness items, half done or unchanged

3. Decoupling: SD socket fixed, headphone amp still bare

(R4-P2-06 · R4-P2-07 · R4-P2-08)

- **Done:** C28 10 µF and C27 100 nF now sit 5.2 / 6.9 mm from the SD socket's VDD pin — the rev-3 "120 mm to the nearest cap" item is closed.
- **Not done:** the optional second 10 µF at Daisy pin 38 (C28 is 27.9 mm from it — acceptable); the TPA6110A2 headphone amp still has only 100 nF on its 3V3_A rail, third revision running (F-031).
- **Potential consequence:** distortion or a faint whine into long headphone leads; the mic amp shares the same under-filtered rail.
- **Fix:** 10 µF (or more) on /+3V3_A at U3.

4. Digital traces beside audio traces, unchanged

(R4-P5-02 · R4-P5-03)

- **What stands:** /USART1_RX runs 49 mm beside AUDIO_IN_L at 0.41 mm, /USART1_TX 26 mm at 0.45 mm; **/TAC_SWITCH_2 runs 27 mm beside AUDIO_OUT_R at 0.20 mm** (and 32 mm beside AUDIO_OUT_L at 0.70 mm) — the closest pairing on the board, same layer.
- **Potential consequence:** a tick on the audio input on MIDI traffic; a click on the line out when the right tactile switch is pressed.
- **Fix:** move the UART and the TAC_SWITCH_2 trace ≥ 1 mm from the audio lines, or put them on the other layer.

5. /RES_SPI floats until firmware drives it

(R4-P3-05 · unchanged from rev 3)

- **What stands:** the OLED reset line is a bare two-node net with no pull-up or RC.
- **Potential consequence:** the display's reset state is undefined from power-up until the firmware configures the pin — usually invisible, occasionally a garbage first frame or a display that needs a second init.
- **Fix:** 10 k from /RES_SPI to /+3V3_OLED.

Everything else in this tier is as rev 3: SPI lines unterminated but electrically short (optional 22–33 Ω or GPIO slew ≤ medium), USB pair length-matched but via-heavy, audio runs long but 0.3 mm — residual judgments, no action required.

Tier 3 — Fab margins (all five Oct-7 rows still NOT DONE)

Item	Measured	Limit	Potential consequence if left	Fix
MK1 annular ring (R4-P1-02)	0.175 mm (1.0 pad / 0.65 drill)	JLCPCB 0.18 absolute, 0.25 recommended	A stricter lot rejects the order, or a thin ring breaks out and the mic pin lifts. Fabbed OK twice so far	MK1 pads 1/2 → 1.1–1.2 mm
USB-C NPTH-to-GND-pad (R4-P1-03)	0.197 mm	0.20 mm	3 µm under: DFM rejection or a drill break-out into the GND pad possible. Fabbed OK twice	Shrink A1/A12/B1/B12 by 0.05 mm
U6 thermal vias (R4-P1-04)	0.25 mm drill / 0.60 pad	Legal at JLC; project rule says 0.30	Extra drill tool; trips the project's own DRC so real errors hide among known ones	Footprint EP vias → 0.30 / 0.70
Silk outlines (R4-P6-02)	0.12 mm stroke; refdes exactly 1.0 / 0.15	0.15 mm line	Outlines print faint or are dropped. Cosmetic	Footprint outline width 0.15
DRC minimums (R4-P1-12)	`min_track_width 0.0`, `min_clearance 0.0` (`.kicad_pro` identical to rev 3)	—	KiCad never warns about sub-minimum copper; the next thin trace ships unnoticed	Board Setup → Constraints: 0.1 / 0.1
0.1 mm copper (R4-P1-01)	74.9 mm, unchanged (U4 fan-out, U6 EP, USB, 5.9 mm /USART1_TX)	0.10 mm, ±20 %	Zero margin where it remains; DRC at JLC minimums is otherwise clean	Widen the USART/USB/3V3_A bits to 0.15 where room exists

Tier 4 — Hygiene

Item	State in rev 4	Potential consequence if left	Fix
ERC flags (R4-P3-06 · R4-P2-11)	14 missing no-connect flags + 4 missing PWR_FLAGs; ERC 51 items, identical to rev 3	A real wiring error hides among 20 known false errors on the next revision	NC flags on P2 SS/SBU pins and J8 3/9/15/18; PWR_FLAG on /DSY_VIN, /+9V_FLAG, Net-(U4-VDD), GND
Hand-assembly sheet (R4-P4-03)	One-sheet BOM; 18 THT parts + the hand-soldered LEDs listed nowhere	A board passes assembly and fails on the bench because a through-hole cap or the LEDs were never fitted	Second BOM sheet with value, footprint, net and source for every hand-fitted part
Position-file flags (R4-P4-04)	"Exclude from position" only on J3–J7; 15 CPL rows with no BOM match	JLC upload shows 15 unmatched entries; harmless but noisy	Set the flag on all THT parts
BOM cell hygiene (R4-P4-05)	Numeric footprint cells, mojibake, one stale cell (D2–D5 say "SOD-323" for the SOD-123FL part ordered)	None at JLC (keys on LCSC #); confuses a human reader	Tidy the cells; fix the D2–D5 text
Refdes repurposed (R4-P4-06)	C19–C22 100 µF → 10 nF, R31/R32 3.3 Ω → 10 k between rev 3 and rev 4	Any rev-3 note, photo or rework instruction naming them now points at a different part	One line in the change log; renumber instead next time
Extended parts (R4-P4-07)	U6, L1, R4, R29/R30, R35–R40, C3339 ×5, D2–D5	Per-line handling fees; R35–R40 is avoidable (C21190 1 % is Basic)	Swap R35–R40
Dangling vias (R4-P1-07)	2, on /LED_2_R and /LED_2_B, unchanged	Two needless holes; cosmetic	Delete
Back-side artwork (R4-P6-01)	3,900 polygons bury 97 of 104 bottom refdes; runs off all edges; nothing on a pad	Inspection and rework harder; assembly unaffected	Refdes halo, or move refdes to Fab
Headphones1 (R4-P3-07 · R4-P6-07, waiting on Trey)	Linear (B) taper as volume control; rotation sense unverified; bushing 9 mm outside the outline	Volume may run backwards or feel wrong; panel needs a cut-out	Trey to confirm; prefer A-taper
SD card detect (R4-P3-09)	/SWA single-pin; all 40 Daisy pins used	Card presence unreadable by firmware	Pin trade if the feature is wanted
SW1 contact code	Not on the JLC BOM, unverifiable	A logic-level (gold 0.4 VA) contact pits at 0.5 A and goes intermittent	Check the ordered part is 1101A4VQEA with silver 5 A contacts

What passed (verified, cited)

- Schematic ↔ PCB 112 / 112 parts, 0 net differences, 0 duplicate refdes.
- BOM ↔ schematic ↔ PCB 92 / 92 three-way; values 92 / 92; the four new LCSC numbers verified by part page; the new bridge part fits its pads.

- U6 fed from the FB6 node at ≈ 8.1 V (3–17 V window); TPS62172 loop unchanged; L1 on the Murata pattern.
- LED currents 11.6 / 12.1 mA, 57–69 mW in 0603; polarity correct for the Würth part; 3.3 V-only Daisy pins behind 1 kΩ bases.
- D6 and FB7 gone; 1N5817W bridge on matching pads; Daisy VIN 6.8–7.5 V; grounds tied; reverse polarity by the bridge; IPC-2221 ampacity on every rail.
- SD socket decoupled; encoder pull-ups and caps wired and routed; encoder clicks routed.
- J8 wiring = NHD serial pin table 20 / 20; 4-wire SPI strap; NC pins correct.
- DRC at JLCPCB minimums: 0 clearance / width / edge / drill errors; courtyards, outline, mask, hole spacing clean.

Question for Trey

- **Headphones1:** is CW = louder as wired, and is a linear taper intended? (carried from rev 3)

Threshold-source caveats

As rev 3: Daisy, Cree and Murata hosts blocked from this session; Daisy pin facts from the KiCad symbol and libDaisy; SSD1322 10 MHz from memory. JLCPCB capabilities and the four new part pages fetched live 2026-10-13.